

The future of service: The power of emotion in human-robot interaction

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ABSTRACT

Astoundingly, recent technological advancements have enabled robots to display emotions. Yet, while emotional expression is valued in the field of service, understanding emotions in human-robot interaction remains underexplored. Since emotions are contagious/transmittable, this study utilised Instagram data to uncover how emotional robots influence potential consumers' affective feelings. By employing machine learning algorithms and sentiment analysis, the findings suggest that the expressions of surprise and happiness are key to creating positive impacts on potential consumers. The cross-disciplinary nature of this study lays the groundwork for next-level social, design, and creative experiences in artificial intelligence research regarding consumer service and experience contexts.

1. Introduction

The integration of artificial intelligence in customer service and experience has been one of the most exciting yet disruptive practices to date. With the ongoing worldwide health crisis, one cannot deny the fact that service automation is shifting the paradigm from high-touch to high-tech (Zeng et al., 2020). The use of robots has increasingly emerged in retail, airports, hotels, restaurants, and transportation areas, amongst others (Prentice and Nguyen, 2020; Zeng et al., 2020). Based on their appearance, robots adopted in service environments can be roughly classified as a combination of functional, cartoonish, and humanlike, or mixed robots (Tung and Au, 2018). With a particular focus on humanoid robots that can anthropomorphise human behaviour most efficiently (Christou et al., 2020), recent studies have investigated how their voice appearance (Yu, 2020b), voice (Murphy et al., 2019), and facial expressions (Yu and Ngan, 2019) act upon consumers during service encounters. In fact, these nonverbal cues are of core importance in any face-to-face interaction since they support and add additional meaning to verbal communication (Pantano, 2020). Among various nonverbal signals, facial expressions form a universal language by presenting emotional states and feelings (Gendron et al., 2018). Through the activation of facial muscles, humans can express emotions including anger, disgust, fear, happiness, sadness, and surprise (Gendron et al., 2018). Notably, emotions are also known to be contagious and can be transmitted to others (Torre et al., 2020). This notion is referred to as

emotional contagion (Hatfield et al., 2014), which has been well-documented in human-human interactions (Rymarczyk et al., 2019; Woo and Chan, 2020). In such an interaction, one tends to subconsciously mimic the facial expression displayed by another individual or interlocutor (Otterbring, 2017). This is especially salient in customer experiences, highlighting direct interactions between consumers and service agents (Söderlund, 2017). In particular, previous studies have underpinned the interplay between emotions and service recovery (Ngan and Yu, 2019), consumer satisfaction (Söderlund, 2017), and customer empathy (Ngo et al., 2020).

Advanced technologies in artificial intelligence and machine learning have revolutionised the nature of consumer experiences by enabling robots to support services. For instance, Hilton Worldwide has employed a service robot, Connie, as the concierge (Lu et al., 2019), while Japan's Kinosaki Onsen hot spring resort has adopted an interactive humanoid robot to communicate with tourists in the information centre (Frommer, 2015). More recently, a Hong Kong-based company developed Sophia, the first anthropomorphic robot capable of displaying more than 60 different emotions (Faraj et al., 2020). Alongside the COVID-19 pandemic, acting as a driving force behind robotisation in the field of general service, businesses have envisioned robots as a means to help facilitate operations during this new period of "normality" (Seyitoglu & Ivanov, 2020). However, apart from the fact that practitioners may still be unprepared for the integration of robotics (Ivanov et al., 2020), service employees also have concerns surrounding potential job

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replacements (Io and Lee, 2020). While some suggest that humanoid robots are preferred by consumers compared to other types (Christou et al., 2020), other findings have revealed contradictory results (Mende et al., 2019).

In that vein, consumers' negative feelings (e.g. fear, anger, and sadness) during human-robot interaction should not be overlooked (Christou et al., 2020) since they could lead to subsequent negative reactions. Yet, the notion of whether emotional contagion can be extended to human-robot interaction remains largely unclear. Since displaying emotions increases the degree of human likeness and stimulates emotional bonds between users and robots (Rincon et al., 2019), it signifies the need to disclose the emotional transmission in human-robot communication (Stock, 2016; Yu and Ngan, 2019). Leveraging consumer emotions in the context of human-robot services is beneficial to value co-creation (Neuhofer et al., 2020b), satisfaction (Yu and Ngan, 2019), and acceptance towards new technologies (Fuentes-Moraleda et al., 2020). Although one can self-report his/her own feelings towards robots (Christou et al., 2020), advanced data analytics has enabled researchers to take a step forward by assessing one's emotions through natural language processing, machine learning (Chatterjee et al., 2019), and facial recognition technology (Yu, 2020a).

As such, this research aims to investigate how robots with emotions influence potential consumers' affective states. By using machine learning algorithms and sentiment analysis techniques based on Instagram data, this study examines the extent to which emotions can influence viewers' feelings. This study is novel in that it bridges and incorporates the disciplines of psychology and data science into the context of human-robot interaction within the domain of service research. Indeed, despite the full potential of robots' emotional capacity still remaining far beyond reach, this research calls for the need to look beyond the status quo and uncover the possible effects of emotional intelligence in relation to foreseeable service ecosystems. By zooming in on what may likely transform the future of experiences, the emotional impact between robots and potential consumers, this study lays the groundwork for service businesses that intend to adopt anthropomorphic and/or humanoid robots. It also offers insights into and opens up several avenues of investigation for the design and application of anthropomorphic robots.

2. Literature review

2.1. Emotional contagion: the what and how

Facial expressions are critical for any form of social communication between humans (Ngan and Yu, 2019), and some emotional expressions, such as anger, contempt, disgust, fear, joy, sadness, and surprise, have been noted across cultures as universal concepts (Gendron et al., 2018). These feelings and emotions serve as the cornerstone of all experiences (Gohary et al., 2016) as they facilitate cognitive processes and help individuals make sense of the world (Isen, 2009). Moreover, human affective states are influential towards one's motives and behaviours (Isen, 2009). One particular reason being that emotions can be transmitted (Torre et al., 2020; Woo and Chan, 2020). In this way, individuals can be infected by someone else's happiness or sadness and vice versa, working as a reciprocal process. Grounded in the discipline of psychology, this phenomenon is referred to as emotional contagion (Hatfield et al., 2014) and reflects the fact that humans, either consciously or unconsciously, mimic the emotion expressed by people around them (Prentice, 2016). Neuroscientists suggest the mirror neuron system as a possible explanation behind the dynamics of emotional contagion (Decety and Ickes, 2011). In fact, transferring emotions not only exists in humans but also occurs in other non-human primates, for example, dogs with their body postures (Palagi et al., 2015).

Broadly speaking, the process of emotional contagion involves three stages; namely, mimicry, feedback, and contagion (Hatfield et al., 2014). As soon as an emotion has been (unconsciously) recognised, one

might begin copying another individual's gestures or facial expressions. This notion of mimicry is often spontaneous and helps humans empathise with others by simulating their feelings (Rymarczyk et al., 2019). Thereafter, the feedback stage initiates, referring to when one starts to truly understand the feeling portrayed by another (Woo and Chan, 2020). As a result of this feedback, the emotion of others is then transformed into one's own experience (Hatfield et al., 2014), and the feeling becomes prominent.

Previous studies have long recognised the importance of this series of actions. For instance, one study highlighted the relationship between the emotions of employees and consumers at the front desk (Ustrov et al., 2016). Similarly, Otterbring (2017) demonstrated that the expression of happiness and joy displayed by employees increases the level of consumer pleasure during service encounters. Nonetheless, given the multifaceted nature of emotions, knowledge related to negative affectivity in service experiences remains under-researched and merits further attention (Liu et al., 2019). Liu and Liu (2020) claim that negative emotions presented by service agents cannot be transmitted to consumers potentially because employees tend to hold back negative facial expressions in service contexts. As such, Liu et al. (2019) discovered that the contagion of negative feelings is more obvious in relation to surface acting, the superficial display of emotions. Notably, however, interactions in contemporary service contexts are no longer limited to humans. Instead, the emergence of service robots, along with anthropomorphic robots as well, underlines the significance of human-robot interaction in the field of service (Choi et al., 2020).

2.2. The role of emotion in human-robot interaction

Essentially, human-robot interaction is considered multidisciplinary in nature, involving the fields of psychology, social sciences, cognitive science, artificial intelligence, design, engineering, and computer science (Joe and Song, 2019). The study of robots in service environments expands the knowledge of consumers' experiences (Neuhofer et al., 2020a) and paves the way for potential applications of service robots (Choi et al., 2020). Notwithstanding the ongoing debates and controversial issues related to safety and security (Ivanov et al., 2020; Mende et al., 2019), some stakeholders in the service industries have envisioned anthropomorphic/humanoid robots as the next contemporary social practice (Choi et al., 2020). On that note, a number of studies in particular have hinted at emotions counting as the most crucial factor in human-robot interaction (Shank et al., 2019; Yu and Ngan, 2019) since they facilitate engagement and co-create experiences (Rincon et al., 2019). Torre et al. (2020), for instance, revealed that consumers present a higher degree of trustworthiness towards robots displaying positive emotions. Likewise, Appel et al. (2020) pointed out that robots with emotions are preferred over a neutral counterpart. This notion is closely related to the theory of uncanny valley, suggesting the relationship between human-likeness and emotions (Appel et al., 2020; Mathur et al., 2020). However, humanoid robots that imperfectly resemble actual human characteristics could provoke negative emotional responses in observers (Mathur et al., 2020). Although the significance of subtle emotional attributes has been established, only more recently have scholars started to investigate the interchange between emotions/-feelings presented by both robots and humans (Mathur et al., 2020; Shank et al., 2019; Torre et al., 2020).

Closely related to robotics, earlier research has underpinned the fact that adding emotional elements to human-computer interactions can enrich conversation (Zhou et al., 2017). If chatbots deliver messages with a joyful tone, users typically experience more positive and happy feelings of emotion (Zhou et al., 2017). Other scholars have also looked into the simple emotions expressed by the Nao robot as front-line service agents, suggesting that emotional contagion is more likely to occur with positive emotions rather than negative ones (Stock, 2016). Yet, Raffard et al. (2016) found opposite results in the context of clinical services in which negative expressions (e.g., anger) were recognised quicker in

comparison to positive emotions (e.g., happiness). These contradictory findings may result from the nature of the research field since users might be more sensitive towards negative emotional expressions within a hospital environment (Raffard et al., 2016). Thus, to further improve trust and cooperation between humans and robots in healthcare services, researchers developed a robot capable of expressing empathy in accordance with users' emotional states. This is achieved through the robot's movements of its eyebrows and mouth to present seven different emotions (Rincon et al., 2019).

Overall, despite existing literature revealing some interesting insights into robots with emotions, the study subjects only resemble the shape of humans (e.g., Nao and Pepper) rather than human physical appearance. With the rise of artificial intelligence, Henn-na hotel in Japan, which used to be fully staffed by robots, has further employed anthropomorphic robots as front desk agents (Ivanov et al., 2018). A study by Yu and Ngan (2019) initiated experimental research by manipulating the smiling behaviour of anthropomorphic robots at Henn-na hotel. In particular, positive emotional displays led to a higher level of perceived service quality (Yu, 2018), interpersonal warmth, and customer satisfaction (Yu and Ngan, 2019). Indeed, although it remains difficult to capture the full potential of robots with emotions given the lack of real interaction, consumers generally expect artificial intelligence to be able to process human emotions so as to provide personalised offers during experiences (Grundner and Neuhofer, 2021).

2.3. A new paradigm in human-robot research

While the importance of emotions and facial expressions has been acknowledged, little research has quantified and determined different emotions in human-robot interaction in a data-driven and systematic way. Derived from the discipline of data science, big data has provided a sea change regarding how knowledge can be constructed and research can be conducted (Balazka and Rodighiero, 2020). In fact, it is the explorative nature of big data that allows researchers to transform unstructured datasets into meaningful and interpretable outcomes (Arora et al., 2019). Emerging data analysis techniques in marketing and management include, but are not limited to, machine learning, artificial intelligence, natural language processing, and data mining. In relation to this particular focus of study, analysing emotions/feelings can best be achieved through the means of natural language processing (e.g., sentiment analysis) when it comes to texts (Celuch, 2020) and facial recognition technology in the case of pictures/videos (Nie, 2020). More specifically, sentiment analysis quantifies the valence of a sentence based on the provided text (Celuch, 2020). Under the guise of natural language processing, detecting the polarity of textual information provides additional value regarding consumers' feelings towards service robots (Fuentes-Moraleda et al., 2020), uncovers users' affective states when communicating with robots (Atzeni and Recupero, 2020), and optimises reactive emotions to improve human-robot interaction (Li et al., 2017).

In addition to text data, facial recognition systems, a method of biometric identification, identify and analyse human faces based on digital images or videos (Nie, 2020; Pantano, 2020). More service businesses have implemented such technology in order to better understand consumer experiences (Pantano, 2020). In human-robot interaction, facial recognition has mainly been installed in robots so as to detect the emotions of users for the purpose of entertainment (Hasegawa and Takahashi, 2019) and security (Nie, 2020). Additionally, with improvements in artificial intelligence, researchers have embraced the potentials of cloud-based services such as the emotion/facial recognition application programming interfaces (API) provided by Microsoft, Google, and Amazon (Pantano, 2020). For instance, Chen et al. (2020) adopted Microsoft Azure to detect the emotional states of consumers based on photographs, while a Japan railway company uses IBM Watson to analyse customers' tone and emotion (Mich, 2020). Regardless of the fact that further investigation on the extent to which

robots can mimic human facial expressions is required (Wirtz et al., 2018), understanding the effect of emotional contagion in human-robot interaction can, without a doubt, add empathy and provide more personalised service experiences in the future (Mich, 2020).

3. Methodology

Overall, this research aims to examine the interplay between the emotional expressions produced by robots and consumers' affective responses using machine learning methods. This study measured robotic emotional expressions in terms of anger, contempt, disgust, fear, happiness, neutral, sadness, and surprise, whereas the consumers' affective reaction was measured by positive, negative, and neutral sentiments. Sophia, the first anthropomorphic robot to express over 60 human feelings/emotions, was selected as the study context (Faraj et al., 2020). Sophia was developed by Hanson Robotics in Hong Kong and was even granted Saudi Arabian citizenship in 2017 (Hanson Robotics, 2020). Technically speaking, Sophia relies on deep learning and advanced material technology to simulate the major facial muscles of humans, allowing it to present joy, confusion, surprise, and sorrow, amongst others. Nonetheless, it is important to note that despite the ultimate goal of Sophia being able to serve in healthcare, customer service, and education (Hanson Robotics, 2020), Sophia has yet to be employed in the service industries. In this research, the methodological procedure involved six distinct steps, which will be presented in more detail below.

3.1. Step 1: Data extraction

To identify pictures of Sophia for the subsequent analysis, this study extracted photos published on Sophia's official Instagram account (@realsophiarobot: <https://www.instagram.com/realsophiarobot/>). Data extraction was conducted in February 2020 using Octoparse, a web scraping tool that has increasingly been applied to service and marketing contexts (Lu et al., 2021; Yu et al., 2020). Based on a total of 391 posts available at that time, the links of posts and images, the type of posts (i.e., pictures/videos), and the captions were crawled. Meanwhile, corresponding comments for each of the posts, timestamps, and usernames were also extracted. This resulted in a total of 8,989 comments left by 4,308 Instagram users.

3.2. Step 2: Data cleaning

To analyse Sophia's emotional expressions, this study focused explicitly on images. Thus, 42 video-posts and their corresponding comments were removed. Thereafter, the researchers manually sifted through the data in order to remove pictures with "cartoonish" characteristics of Sophia (e.g., Little Sophia: https://www.instagram.com/p/BtPD_iFHL5H/). Next, the remaining posts were imported into a facial recognition API developed by Microsoft Azure. This is an artificial intelligence service that detects the faces and attributes of an image (e.g., Chen et al., 2020) and returns image annotation, including a range of facial expressions. Pictures in which a face was not recognised were excluded. This may have occurred due to Sophia being too small or looking back in the pictures. Therefore, the cleaned dataset resulted in 223 posts with 6,041 comments left by 4,308 Instagram users.

3.3. Step 3: Emotional recognition of Sophia

Sophia's emotional expressions were identified by the facial recognition API, as introduced in the previous section. It is important to note that in the pictures containing several people, only Sophia's emotions were detected. The links to the pictures were first imported to Microsoft Azure, which returned the facial attributes of anger, contempt, disgust, fear, happiness, neutral, sadness, and surprise. The score, ranging from 0 to 1, implies the strength of a specific emotion. In this case, a higher

value indicates a stronger emotional expression displayed in the picture.

3.4. Step 4: text pre-processing

The 6,041 comments identified in step 2 were pre-processed for the following sentiment analysis so as to uncover consumers' affective reactions. First, non-English comments and unknown characters, URLs, numbers, and references to usernames (i.e., @username) were removed. Additionally, words followed by a hashtag (#) were eliminated as well. Next, the researchers prepared a list of stopwords to remove irrelevant terms (e.g., 'and', 'you', 'I'). These procedures resulted in a total of 3,777 remaining comments, which were then transformed into lowercase. In order to reduce morphological variation, comments were lemmatised and stemmed using Wordnet Lemmatizer and Porter Stemmer, respectively. Finally, comments were tokenised into small units for the application of sentiment analysis.

3.5. Step 5: Sentiment analysis

To detect the affective states embedded in the comments, this study employed a lexicon-based sentiment analysis with the incorporation of the Vader algorithm. The Vader algorithm was chosen based on its advantages in analysing emojis, slangs, and acronyms commonly used by social media users (Celuch, 2020). It also outperforms other sentiment analysis techniques for social media contexts because Vader is built on a valence-based sentiment lexicon in accordance with Natural Language Toolkit (Gilbert and Hutto, 2014). Technically speaking, Vader returns four types of sentiments; namely, positive, negative, neutral, and compound values. The compound score suggests a single measure of polarity, ranging from -1 (the most negative) to $+1$ (the most positive), whereas a neutral feeling equates to 0. The compound value, on the other hand, is a normalised score of the sum of computed valence values based on the sentiment lexicon (Costa et al., 2019). Similar to other studies implying its usefulness (e.g., Costa et al., 2019; Lundqvist et al., 2020), the following analysis was also based on the compound score.

3.6. Step 6: Implementation of machine learning

Lastly, in order to analyse how Sophia's emotional expressions act on consumers, machine learning techniques were implemented using the *Auto Model Tool* available in *RapidMiner*. *RapidMiner* is an open-source platform for data mining, machine learning, and predictive modelling, and has been previously applied in research relating to consumer experiences (Buzova et al., 2019; Yan et al., 2018). Overall, on account of this study's datasets containing nonlinear structure, machine learning methods are capable of detecting possible nonlinear relationships and higher-order interactions (Ryo and Rillig, 2017). Distinct from conventional statistical modelling (e.g., regression analysis), machine learning algorithms stand out on the premise that prior assumptions are not required for the datasets (Recknagel, 2001).

In order to ensure the reliability of the results, the relative performance of four different methods commonly used in supervised learning for regression (i.e., decision tree, random forest, gradient boosted trees, and support vector machine) were compared. The first three techniques entail tree-based algorithms (Fan et al., 2018). They are non-parametric in nature and are less affected by outliers and multicollinearity. In this regard, decision trees involve a series of sequential steps and produce a simple decision-making diagram (Ogutu et al., 2011). Random forests combine several decision trees at the end of the process, whereas gradient boosting combines trees at the beginning of the process (Ogutu et al., 2011). Contrarily, support vector machine employs distance-based algorithms and is more effective and accurate when a clear margin of separation is present (Fan et al., 2018).

First, the compound sentiment scores, described in section 3.5, were assigned as the target variables. Subsequently, feature selection of the input variables was conducted based on emotional expressions that

could contribute the most to the quality of the machine learning models. In order to find the best features for machine learning (Yu and Liu, 2003), surprise, happiness, neutral, and sadness were included in the learning process. The other emotions (fear, disgust, contempt, and anger) were excluded since nearly all values were identical (i.e., closer to 0). Subsequently, this study made use of the automatic optimisation feature in *RapidMiner*. This algorithm optimises (1) the maximal depth for decision tree, (2) the number of trees and maximal depth for random forest, (3) the number of trees, maximal depth, and learning rate for gradient boosted trees, and (4) the gamma and C hyperparameter for support vector machine.

Similar to recent studies in the field of management/marketing (e.g., Biyik and Tanyeri, 2018; Nikou et al., 2019; Shafqat and Byun, 2020), this research evaluated the performance of the models based on the mean squared error (MSE) and the root mean squared error (RMSE). The former is suitable for datasets with outliers and unexpected values, while the latter is useful for datasets containing large errors (Willmott, 1982). It is relevant to bear in mind that although there is no rule of thumb, a lower value technically indicates better performance. Finally, the algorithms highlighted the most influential emotional features towards the compound sentiment based on weight vectors (Ribeiro et al., 2016). It then visualised and predicted the strength of the different emotions that could provide the best learning models.

4. Results

The facial recognition API categorised one's emotional expressions into seven different states: anger, contempt, disgust, fear, happiness, neutral, sadness, and surprise. Table 1 provides an overview of the intensity of these emotions. Based on Instagram pictures, Sophia's emotions are dominated by neutral and happiness, followed by sadness and surprise. Regarding the valence of the comments in response to Sophia's posts, the findings suggest that Instagram users' feelings towards Sophia have a relatively low value. Furthermore, based on the overall compound scores, a relatively low positive experience towards Sophia was found, with a sentiment value of 0.2 (Table 2). Thus, this opens up an interesting avenue for investigation regarding how specific emotions can be transmitted in human-robot interactions.

Four machine learning algorithms were performed in this study to examine the relationship between robot and human emotions. This study took MSE and RMSE as the performance metrics for evaluation, which are provided in Table 3. Consistent with previous literature (Ebrahimi et al., 2019; Gong et al., 2019), the results demonstrate a higher accuracy score of gradient boosted trees. To be precise, the optimal parameters of the gradient boosted trees identified in this study consisted of 15 for number of trees, 2 for maximal depth, and 0.100 for learning rate. Consequently, the gradient boosting results were chosen for the subsequent data interpretation. Nonetheless, bearing in mind that each algorithm has its pros and cons, one of the epistemological challenges of big data research involves the accuracy of the algorithms as well as the interpretability of the results (Elragal and Klischewski, 2017). Although this study selected the algorithm with the best performance, the reality of big data, meaning that knowledge production and interpretation are often subject to human bias, needs to be taken into account (Balazka and Rodighiero, 2020; Symons and Alvarado, 2016). While acknowledging that epistemological pitfalls warrant further investigation, these are considered beyond the scope of this study.

The learning model produced by the gradient boosting algorithm returned the local weights of Sophia's emotional expressions (see Table 4). These weights in particular suggest which emotions are generally most influential in relation to the valence of Instagram comments. The results demonstrate that surprise is the most prominent input feature affecting Instagram users' affective reactions towards Sophia's images. This is followed by the expressions of happiness, neutral, and sadness.

Table 1
Overview of emotional expressions.

	Anger	Contempt	Disgust	Fear	Happiness	Neutral	Sadness	Surprise
Mean	0.001	0.003	0.001	0.010	0.325	0.513	0.079	0.071
SD	0.002	0.006	0.002	0.034	0.363	0.361	0.147	0.164

Note: the strength ranges from 0 to 1.

Table 2
Overview of sentiment from Instagram comments.

	Positive	Negative	Neutral	Compound
Mean	0.254	- 0.051	0.695	0.227
SD	0.309	- 0.145	0.313	0.387

Note: the sentiment ranges from -1 to +1.

Table 3
Performance of machine learning.

Techniques	Mean squared error	Root mean squared error
Decision tree	0.156	0.395
Random forest	0.155	0.394
Gradient boosted trees	0.154	0.393
Support vector machine	0.176	0.419

Note: the metrics range from 0 to ∞.

Table 4
Local weights of emotional expressions.

Attributes	Weight
Surprise	0.156
Happiness	0.088
Neutral	0.066
Sadness	0.027

Note: weights range from 0 to 1.

In addition, this study provides the optimal combination of inputs required to maximise the forecast of the learning model. The technique is known as prescriptive analytics, aiming to uncover the best course of action (Bertsimas and Kallus, 2020). In essence, it reverses the learning procedure by defining a desired output first and refining input features accordingly. To ensure that only sensible and feasible solutions are created, the constraints of the input features were defined to stay within 2 standard deviations of the average value and to remain above/below its minimum/maximum value. Based on the optimisation algorithm, Fig. 1 visualises and predicts the optimal combination of Sophia's emotional expressions needed to reach the most positive sentiment

score, valued at 0.571.

5. Discussion

As the service industry goes hand in hand with technological improvements, it is critical for scholars and practitioners to maintain a futuristic mindset in order to envision and predict emerging practices in service environments as well as advance and picture promising consumer experiences. Although service robots with emotional intelligence are continuing to progress and emerge, this is merely to a limited extent. Nonetheless, Grunder and Neuhofer (2021) highlight the necessity to go beyond the status quo and look towards the future potential of artificial intelligence, which has been proclaimed to trigger a digital transformation in customer experiences. Automation technologies are already gradually being adopted as a means to overcome the current challenges related to social distancing and the changes in consumer behaviour (Seyitoğlu & Ivanov, 2020). Despite robots having the potential to revitalise the service industries (Auliawan and Ong, 2020), emotions are still inevitable elements in service encounters (Ngan and Yu, 2019; Woo and Chan, 2020). For that reason, this research intertwines the lens of emotional contagion with human-robot interactions.

On a more general level, this research first indicates that although the emotional robot, Sophia, can emulate a wide range of human facial expressions (Faraj et al., 2020), its images presented on Instagram fail to sufficiently demonstrate the feelings of fear, contempt, anger, and disgust, making it impossible to train the machine learning models. In fact, since Instagram allows users to reach a wide range of audiences, these findings are not surprising and can be explained from the perspective of impression management (Hong et al., 2020). Rooted in the discipline of psychology, impression management refers to the (sub) conscious processes in which people control how they are perceived by others during social interaction (Hong et al., 2020). For example, in the services industries, employees can purposely include or exclude certain traits so as to leave a lasting impression on consumers. Keeping this notion in mind, while the pictures used for this study were not posted by Sophia itself, it could be the case that the robotics company prefers to share the "ideal" version of Sophia. This assumption can be proven further considering the dominated, and mostly positive, emotions presented by Sophia (e.g., happiness) on Instagram.

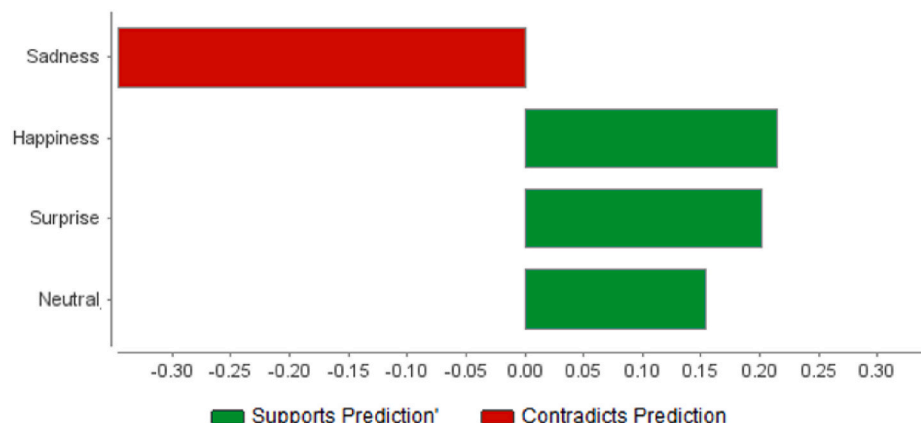


Fig. 1. Optimisation of machine learning model.

However, this study implies that Instagram users tend to use negative terms/phrases when leaving a comment on Sophia's posts. These findings are rather inconsistent with earlier studies in terms of online users' overall feelings towards robots. Even though several researchers claim anthropomorphic robots to be more preferable (Christou et al., 2020; Ivanov et al., 2018), a study conducted by Yu (2020b), investigating the humanoid robot at Henn-na hotel, suggested that robots with an extremely high level of human-likeness can potentially lead to a sense of discomfort. Likewise, another recent study assessing hotel service robots with different degrees of human-likeness concluded that users are less likely to accept highly anthropomorphic robots due to safety concerns, the imitation of human behaviour, and the capabilities of robots to handle multiple tasks (Jia et al., 2021). Furthermore, the explorative nature of this research poses a question of whether potential consumers are able to make accurate inferences about how they would interact with or react to an emotional robot in a real-life situation. As explained by Grunder and Neuhofer (2021), insufficient interaction might evoke hesitation for consumers since they are unsure of a robot's capability to truly understand their feelings and emotions.

When looking at the interplay of different emotions and Instagram users' feelings, this research reveals an interesting result. In particular, the findings suggest that the expression of surprise comes across as the most influential emotion to the observers. Notably, surprise is a complex feeling and might carry any valence (i.e., neutral, negative, or positive) (Noordewier and Breugelmans, 2013). Hence, observers' affective states vary according to the nature of the surprising stimulus (Noordewier and Breugelmans, 2013). Albeit the fact that Sophia presents positive surprise, Noordewier and van Dijk (2019) indicate that observers may still experience a state of interruption before they can truly reflect on their feeling towards the stimulus. Logically, due to the lack of social interaction (Grundner and Neuhofer, 2021) with Sophia, observers' emotional reactions might stay at the initial phase without thinking or processing any further (Noordewier and van Dijk, 2019). Another plausible reason could be that since surprise is often related to a freezing/suspension-like behaviour, viewers might present passive and defensive reactions (Scherer et al., 2004) towards Sophia.

From a human-likeness perspective, the results of this study could also be explained in terms of the theory of uncanny valley (Appel et al., 2020; Mathur et al., 2020). Overall, even when Sophia portrays the feeling of happiness, the polarity of the Instagram comments leans more towards the negative side of the spectrum. Seeing that Sophia is the first anthropomorphic robot to display various emotions, it could be that online users are either not used to robots highly resembling humans (Yu, 2020b) or that the effectiveness and authenticity of emotions presented by robots are still questionable (Wirtz et al., 2018). Furthermore, this study implies that the expression of sadness presents the lowest contribution to the valence of comments. Returning to the topic of emotional regulation, since, assumingly, people are more likely to hinder and hide their negative feelings (Liu and Liu, 2020), the display of sadness may generate a lower chance of transmission to Instagram users as a result.

Finally, while this study places a particular focus on facial expressions and emotional reactions, emotions, per se, might not necessarily lead to Instagram viewers' positive or negative responses. However, this study underlines the display of emotions as one of the subtle yet compelling factors that could imaginably make a difference in the emerging theme of human-anthropomorphic robot interaction.

6. Conclusion

6.1. Theoretical contribution

Among various nonverbal cues, emotions enable individuals to build deeper and more meaningful relationships (Otterbring, 2017; Woo and Chan, 2020). Alongside the emergence of robots, designing experiences that touch humans emotionally is of the utmost importance for researchers and practitioners in any service environment (Neuhofer et al.,

2020a). The development of robots with emotional intelligence additionally brings a major paradigm shift to how individuals interact with robots. By learning from the most advanced human-like emotional robot, Sophia, this study crosses disciplinary boundaries as it connects psychology and data science and provides cutting-edge perceptions of robotics and artificial intelligence research in the context of service experiences. As humans often personify computers/machines (McGinn et al., 2018), this study uses a modern approach of applying social media (i.e., Instagram) data in order to advance the current understanding of human-robot relationships. In addition, this research goes beyond the traditional scope of a well-researched human social phenomenon by highlighting the possibility of emotional transmission, or emotional contagion, among any "organism".

Methodologically speaking, the explorative nature of this study uncovers observations (e.g., expression of surprise) that may have been more difficult to recognise otherwise. It leverages advanced analytics so as to open up new avenues of research and practical applications. Resting on the concept of emotional contagion, the findings zoom in on the interplay between a robot's emotional display and a consumer's potential reactions during service encounters in the near future. Although this research is incapable of providing an exact answer to the question "when will human-level AI become a reality?" posed by Grunder and Neuhofer (2021), this study calls for a need to further investigate robots with emotions. As a value-inducing technology for co-creating service experiences, examining robotics within the contexts of consumer behaviour, design research, and psychological studies should be viewed as a relevant priority in academia.

6.2. Managerial implications

This research presents several implications for service businesses that intend to employ robots with emotional intelligence as well as for robotics companies to optimise the design of human-like robots. First and foremost, the findings imply that the portrayal of the emotion surprise requires supplementary adjustments. More specifically, as surprise can be interpreted as either positive or negative, it is critical to re-evaluate what kind of messages service robots intend to deliver. Technicians and designers might need to adjust the intensity of the robot's facial muscles (Lucey et al., 2010) triggering surprise; namely, the inner and outer brow raisers, the upper lid raiser, the lips part, and the mouth stretch. In the same vein, it could be important for technicians to test the optimal intensity in relation to the cheek raisers, the lip corner pullers, and the lips part for the expression of happiness (Lucey et al., 2010). Despite the fact that, based on the comments of the pictures, observers did not present a very positive attitude towards Sophia, the results provide robotics companies with guidelines to optimise observers' emotional reactions. In particular, this study suggests that the combination of the facial expressions happiness and surprise can reach the best outcomes (see Fig. 1). On that note, the display of positive surprise is more likely to prevent viewers from having negative feelings towards Sophia. In addition, this research also indicates that the activation of facial muscles for sadness (i.e., inner brow raiser, outer brow raiser, brow lowerer, lip corner depressor, and chin raiser) (Lucey et al., 2010) should be minimised so as to avoid any negative reactions and impressions.

6.3. Limitations and recommendations for future research

This study is not without its limitations. Primarily, it is critical to note that although the service industries, including retail, hospitality, and healthcare, amongst others, could benefit from emotional robots, Sophia has yet to be employed and implemented into real-life conditions. As such, it is also necessary to bear in mind that the Instagram commenters (referred to as potential consumers) in this study did not personally interact with any robots. Thus, since Sophia has not been used as a service robot, Instagram users may have a slightly different

mindset compared to consumers. Instead, the current practice seems to be much closer to previous research in which vignettes were applied in order to examine human-robot interaction (e.g., Yu and Ngan, 2019). However, the more urgent and important message is to focus on envisioning the power of emotional expression and encouraging researchers to be prepared for the emerging practices in future human-robot interaction and communication.

Meanwhile, as feelings based on Instagram comments do not fully represent the internal states of consumers, it is recommended to consider experimental design and to incorporate other biometrics, such as eye-tracking, electroencephalography, or galvanic skin response, in future studies. Scholars could also apply other text analytics such as topic modelling to disclose the content of comments together with the sentiment of texts. Furthermore, facial recognition algorithms are often trained on exaggerated human facial expressions, questioning to what extent machine learning can truly recognise emotions presented by human-like robots. Likewise, while this research assumed that big data methods could be leveraged in this area, some of the common emotional expressions were not viable due to the fact that these expressions ultimately did not contribute to the machine learning model. To address this issue, scholars are urged to involve human-raters to supplement and validate the machine learning categorisation results. Finally, one should note that the current study's findings are only representative in the context of Instagram. Once the development of robots with emotional intelligence becomes more mature in the future, scholars are suggested to continue this line of research by investigating whether emotional contagion still applies in human-robot interaction.

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